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```python
LAST_NAME = "Cajida"
STUDENT_ID = "TUPM-26-1243"
FAVORITE_ARTIST = "TUKI."

SEED_DIGIT = int(STUDENT_ID[-1])
ID_SUM = sum(int(d) for d in STUDENT_ID if d.isdigit())
NAME_LENGTH = len(LAST_NAME)
ARTIST_LENGTH = len(FAVORITE_ARTIST)
ARTIST_SUM = sum(ord(c) - 64 for c in FAVORITE_ARTIST.upper() if c.isalpha())

print("Seed:", SEED_DIGIT)
print("ID Sum:", ID_SUM)
print("Name Length:", NAME_LENGTH)
print("Artist Length:", ARTIST_LENGTH)
print("Artist Sum:", ARTIST_SUM)
```

[1] Python

```
Seed: 3
ID Sum: 18
```

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```
ID Sum: 18
Name Length: 6
Artist Length: 5
Artist Sum: 61
```

```python
record_count = 10 + ARTIST_LENGTH
original = [ID_SUM + i * (NAME_LENGTH + 1) for i in range(record_count)]
rotation = SEED_DIGIT % len(original)
rotated = original[rotation:] + original[:rotation]
target = original[(ID_SUM + ARTIST_SUM) % len(original)]

print("Original Dataset:", original)
print("Rotation Point:", rotation)
print("Rotated Dataset:", rotated)
print("Target Incident:", target)
```

[2] Python

```
Original Dataset: [18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95, 102, 109, 116]
Rotation Point: 3
```

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```
Rotated Dataset: [39, 46, 53, 60, 67, 74, 81, 88, 95, 102, 109, 116, 18, 25, 32]
Target Incident: 46
```

```python
def rotated_search(data, target):
    low, high, comparisons = 0, len(data) - 1, 0
    while low <= high:
        mid = (low + high) // 2
        comparisons += 1
        if data[mid] == target:
            return mid, comparisons
        if data[low] <= data[mid]:
            if data[low] <= target < data[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if data[mid] < target <= data[high]:
                low = mid + 1
            else:
                high = mid - 1
    return -1, comparisons

def linear_search(data, target):
    for i, value in enumerate(data):
        if value == target:
            return i, i + 1
    return -1, len(data)

ri, rc = rotated_search(rotated, target)
li, lc = linear_search(rotated, target)

if ri == -1:
    print("Target incident not found")
else:
    print("Target found at index", ri)
print("Rotated Binary Search comparisons:", rc)
print("Linear Search comparisons:", lc)
```

[3] Python

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```python
else:
    print("Target found at index", ri)
print("Rotated Binary Search comparisons:", rc)
print("Linear Search comparisons:", lc)
print("Time Complexity: Rotated Binary Search O(log n), Linear Search O(n)")
print("Space Complexity: O(1)")
```

[3] Python

```
Target found at index 1
Rotated Binary Search comparisons: 3
Linear Search comparisons: 2
Time Complexity: Rotated Binary Search O(log n), Linear Search O(n)
Space Complexity: O(1)
```